Dorea Fernando - Emanuel
grupa 143

Conice

Definiție

Conicele sunt figuri (curbe) ce se obțin prin intersectarea unui plan și a unui con.

Fie forma patratică afină:

$$f(x,y) = a_{11}x^2 + 2a_{12}xy + a_{22}y^2 + 2b_1x + 2b_2y + c$$

Mulțimea $T = \{ M(x,y) \mid (x,y) \in \mathbb{R}^2,\ f(x,y) = 0 \}$ se numește conică algebrică de ordin al doilea

$$a = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} \qquad A = \begin{pmatrix} a_{11} & a_{12} & b_1 \\ a_{12} & a_{22} & b_2 \\ b_1 & b_2 & c \end{pmatrix}$$

$$\delta = \det a$$

$$\Delta = \det A$$

$$\Delta_1 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{11} & b_1 \\ b_1 & c \end{vmatrix} + \begin{vmatrix} a_{22} & b_2 \\ b_2 & c \end{vmatrix}$$

Clasificare conice

δ	Δ	$I\Delta$	Δ_1	Conică	Gen
>0	$\neq 0$	<0		elipsă	eliptic
	$\neq 0$	>0		conica vidă	
	$=0$			punct dublu	
<0	$\neq 0$			hiperbolă	hiperbolic
	$=0$			drepte concurente	
$=0$	$\neq 0$			parabolă	parabolic
	$=0$		<0	drepte paralele	
	$=0$		$=0$	drepte duble	
	$=0$		>0	conică vidă	

1. <u>Elipsa</u>

<u>ecuația canonică</u> :

$$\frac{(x^1)^2}{(a_1)^2} + \frac{(x^2)^2}{(a_2)^2} = 1$$

$$q(x) = \frac{(x^1)^2}{(a_1)^2} + \frac{(x^2)^2}{(a_2)^2} - 1$$

$$a = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 \\ 0 & \frac{1}{(a_2)^2} \end{pmatrix}$$

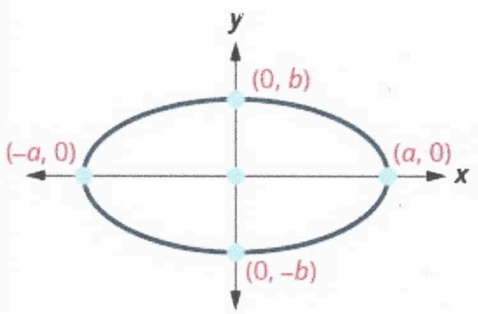
$$A = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

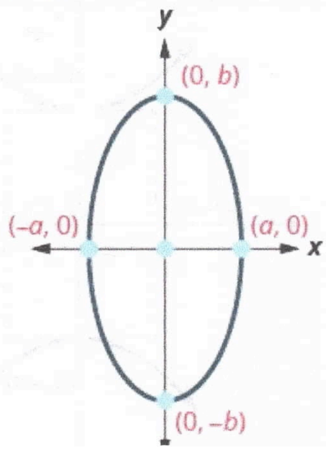
$$\delta = \frac{1}{(a_1)^2 \cdot (a_2)^2}$$

$$\Delta = -\delta$$

$$I = \frac{1}{(a_1)^2} + \frac{1}{(a_2)^2}$$

$$\lambda_1 = \frac{1}{(a_1)^2} \qquad \lambda_2 = \frac{1}{(a_2)^2}$$





2. Hiperbolă

ecuaţie canonică:

$$\frac{(x^1)^2}{(a_1)^2} - \frac{(x^2)^2}{(a_2)^2} = 1$$

$$q(x) = \frac{(x^1)^2}{(a_1)^2} - \frac{(x^2)^2}{(a_2)^2} - 1 = 0$$

$$a = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 \\ 0 & \frac{-1}{(a_2)^2} \end{pmatrix}$$

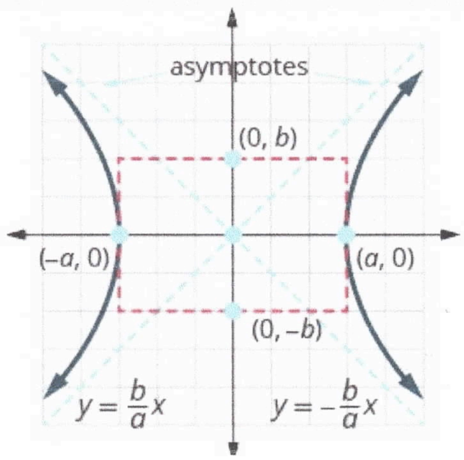
$$A = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{-1}{(a_2)^2} & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

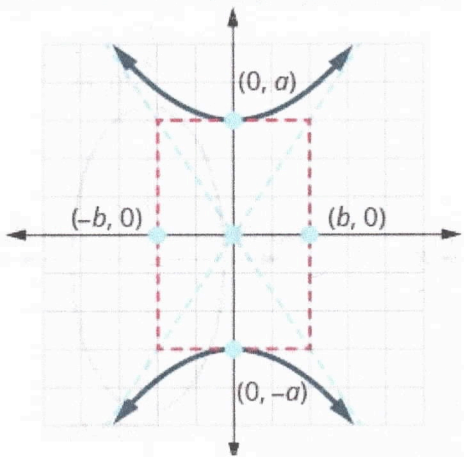
$$\delta = \frac{-1}{(a_1)^2 (a_2)^2}$$

$$\Delta = -\delta = \frac{1}{(a_1)^2 (a_2)^2}$$

$$I = tr(a) = \frac{1}{(a_1)^2} - \frac{1}{(a_2)^2}$$

$$\lambda_1 = \frac{1}{(a_1)^2} \qquad \lambda_2 = -\frac{1}{(a_2)^2}$$





3. Parabolă

ecuație canonică

$$y^2 = 4a_1 x$$

$$g(x) = y^2 - 4a_1 x = 0$$

$$a = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

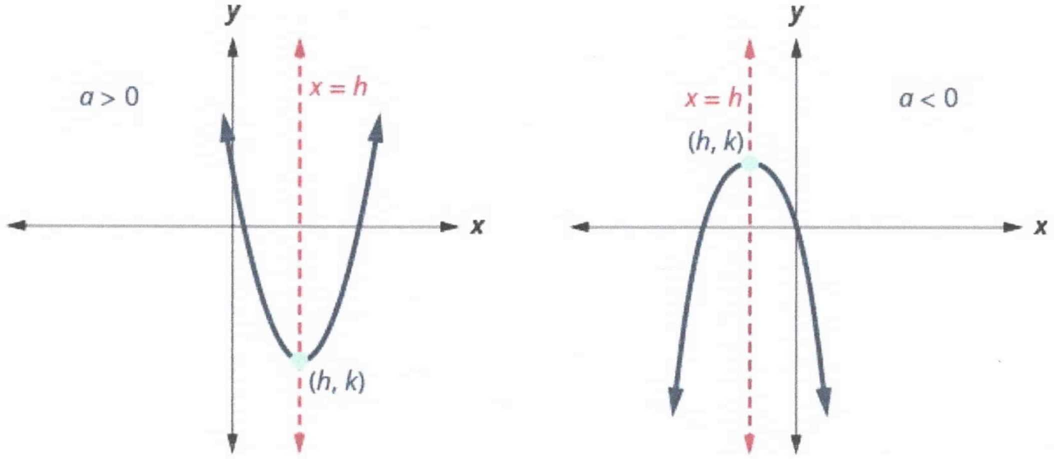
$$A = \begin{pmatrix} 0 & 0 & -2a_1 \\ 0 & 1 & 0 \\ -2a_1 & 0 & 0 \end{pmatrix}$$

$\delta = 0$

$\Delta = 4a_1^2$

$I = 1$

$\lambda_1 = 0 \quad \lambda_2 = 1$



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Cuadrice

Definiție

Cuadricele sunt suprafețe algebrice de gradul al doilea, adică suprafețe ale spațiului afin euclidian tridimensional și reprezintă mulțimea soluțiilor unor ecuații de forma: $a_{11}x^2 + a_{22}y^2 + a_{33}z^2 + 2a_{12}xy + 2a_{13}xz + 2a_{23}yz + 2a_{10}x + 2a_{20}y + 2a_{30}z + a_{00}$.

Invarianți

$$a = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \qquad A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{10} \\ a_{12} & a_{22} & a_{23} & a_{20} \\ a_{13} & a_{23} & a_{33} & a_{30} \\ a_{10} & a_{20} & a_{30} & a_{00} \end{pmatrix}$$

$\delta = \det a$

$\Delta = \det A$

J = suma minorilor diagonali de ordin 2 din a

I = suma minorilor diagonali de ordin 2 din A

K = suma minorilor diagonali de ordin 3 din A

Clasificare

Nr	δ	Δ	$\lambda_1, \lambda_2, \lambda_3$	$\frac{-\Delta}{\lambda_1\delta}, \frac{-\Delta}{\lambda_2\delta}, \frac{-\Delta}{\lambda_3\delta}$	K	L	Cuadrica
1	>0	$\neq 0$		$+ \quad + \quad +$			elipsoid
2	<0	$\neq 0$		$+ \quad + \quad -$			hiperboloid cu o pânză
3	>0	$\neq 0$		$- \quad - \quad +$			hiperboloid cu două pânze
4	<0	$\neq 0$		$- \quad - \quad -$			cuadrică vidă
5	$\neq 0$	0	aceleași semn				punct dublu
6	$\neq 0$	0	$+ \quad + \quad -$				con pătratic
7	0	$\neq 0$	$\lambda_1, \lambda_2 > 0, \lambda_3 = 0$				paraboloid eliptic
8	0	$\neq 0$	$\lambda_1 \lambda_2 < 0, \lambda_3 = 0$				paraboloid hiperbolic
9	0	0	$\lambda_1 > 0, \lambda_2 > 0, \lambda_3 = 0$		>0		cuadrică vidă
10	0	0	$\lambda_1 \cdot \lambda_2 > 0, \lambda_3 = 0$		0		dreaptă dublă
11	0	0	$\lambda_1 > 0, \lambda_2 > 0, \lambda_3 = 0$		<0		cilindru eliptic
13	0	0	$\lambda_1 < 0, \lambda_2 < 0, \lambda_3 = 0$		<0		cuadrică vidă
12	0	0	$\lambda_1 < 0, \lambda_2 < 0, \lambda_3 = 0$		>0		cilindru eliptic
14	0	0	$\lambda_1 \lambda_2 < 0, \lambda_3 = 0$		$\neq 0$		cilindru hiperbolic
15	$\cancel{0}$	$\cancel{0}$					cuadrică vidă
16	0	0	$\lambda_1 \lambda_2 < 0, \lambda_3 = 0$		0		plane secante
17	0	0	$\lambda_1 \neq 0, \lambda_2 = \lambda_3 = 0$		0	>0	cuadrică vidă
18	0	0	$\lambda_1 \neq 0, \lambda_2 = \lambda_3 = 0$		0	0	plan dublu
19	0	0	$\lambda_1 \neq 0, \lambda_2 = \lambda_3 = 0$		0	<0	plane paralele
19	0	0	$\lambda_1 \neq 0, \lambda_2 = \lambda_3 = 0$		$\neq 0$		cilindru parabolic

1. Elipsoid

ecuație canonică: $\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} + \frac{z^2}{a_3^2} = 1$

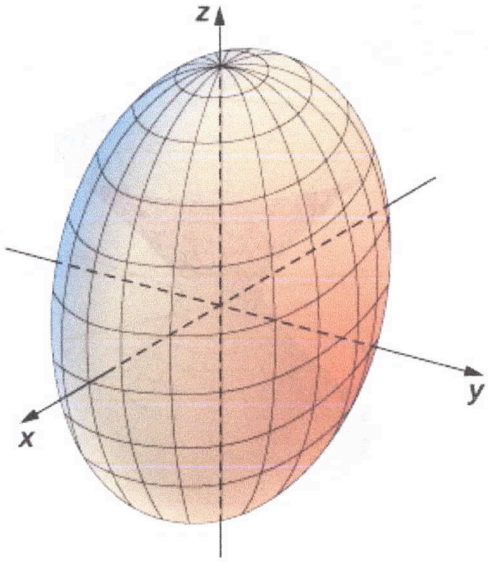
$$q(x) = \frac{x^2}{(a_1)^2} + \frac{y^2}{(a_2)^2} + \frac{z^2}{(a_3)^2} - 1 = 0$$

$$a = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 \\ 0 & 0 & \frac{1}{(a_3)^2} \end{pmatrix} \qquad \delta = \frac{1}{(a_1)^2 (a_2)^2 (a_3)^2}$$

$$A = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 & 0 \\ 0 & 0 & \frac{1}{(a_3)^2} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \qquad \Delta = \frac{-1}{(a_1)^2 (a_2)^2 (a_3)^2}$$

$$\lambda_1 = \frac{1}{(a_1)^2} \quad \lambda_2 = \frac{1}{(a_2)^2} \quad \lambda_3 = \frac{1}{(a_3)^2}$$

$$I = \frac{1}{(a_1)^2} + \frac{1}{(a_2)^2} + \frac{1}{(a_3)^2}$$



2. Hiperboloid cu o pânză

ecuație canonică: $\dfrac{x^2}{(a_1)^2} + \dfrac{y^2}{(a_2)^2} + \dfrac{z^2}{(a_3)^2} = 1$

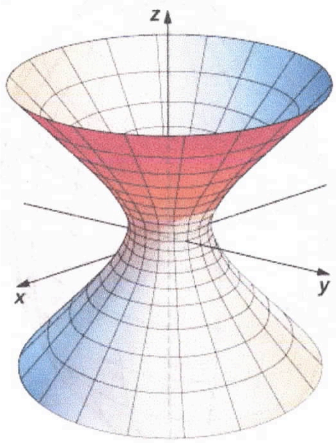
$$q(x) = \frac{x^2}{(a_1)^2} + \frac{y^2}{(a_2)^2} - \frac{z^2}{(a_3)^2} - 1 = 0$$

$$a = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 \\ 0 & 0 & \frac{-1}{(a_3)^2} \end{pmatrix} \qquad \delta = \frac{-1}{(a_1)^2 (a_2)^2 (a_3)^2}$$

$$A = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 & 0 \\ 0 & 0 & \frac{-1}{(a_3)^2} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \qquad \Delta = \frac{1}{(a_1)^2 (a_2)^2 (a_3)^2}$$

$$\lambda_1 = \frac{1}{(a_1)^2} \qquad \lambda_2 = \frac{1}{(a_2)^2} \qquad \lambda_3 = \frac{-1}{(a_3)^2}$$

$$I = \frac{1}{(a_1)^2} + \frac{1}{(a_2)^2} - \frac{1}{(a_3)^2}$$



3. Hiperboloid cu două pânze

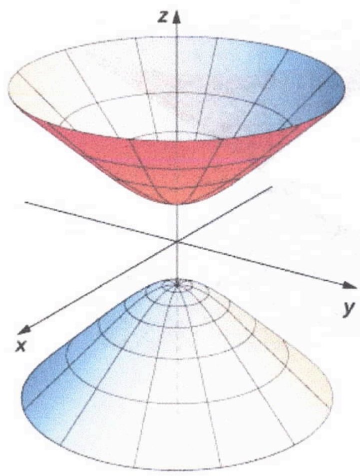
ec. canonică: $\dfrac{z^2}{(a_3)^2} - \dfrac{x^2}{(a_1)^2} - \dfrac{y^2}{(a_2)^2} = 1$

$$g(x) = -\frac{x^2}{(a_1)^2} - \frac{y^2}{(a_2)^2} + \frac{z^2}{(a_3)^2} - 1 = 0$$

$$a = \begin{pmatrix} \frac{-1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{-1}{(a_2)^2} & 0 \\ 0 & 0 & \frac{1}{(a_3)^2} \end{pmatrix} \qquad \delta = \frac{1}{(a_1)^2(a_2)^2(a_3)^2}$$

$$A = \begin{pmatrix} \frac{-1}{(a_1)^2} & 0 & 0 & 0 \\ 0 & \frac{-1}{(a_2)^2} & 0 & 0 \\ 0 & 0 & \frac{1}{(a_3)^2} & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \qquad \Delta = \frac{-1}{(a_1)^2(a_2)^2(a_3)^2}$$

$$\lambda_1 = \frac{-1}{(a_1)^2} \qquad \lambda_2 = \frac{-1}{(a_2)^2} \qquad \lambda_3 = \frac{1}{(a_3)^2}$$



4. <u>Con eliptic</u>

ec. canonica : $\frac{x^2}{(a_1)^2} + \frac{y^2}{(a_2)^2} - \frac{z^2}{(a_3)^2} = 0$

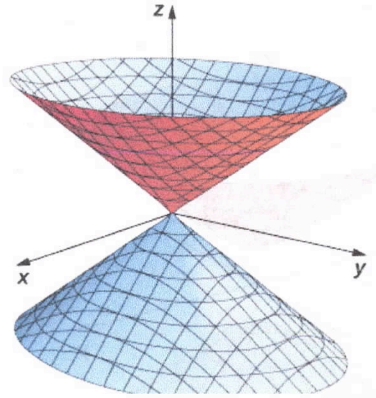
$$g(x) = \frac{x^2}{(a_1)^2} + \frac{y^2}{(a_2)^2} - \frac{z^2}{(a_3)^2} = 0$$

$$a = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 \\ 0 & 0 & \frac{-1}{(a_3)^2} \end{pmatrix} \qquad \delta = \frac{-1}{(a_1)^2(a_2)^2(a_3)^2}$$

$$A = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 & 0 \\ 0 & 0 & \frac{-1}{(a_3)^2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \Delta = 0$$

$$\lambda_1 = \frac{1}{(a_1)^2} \qquad \lambda_2 = \frac{1}{(a_2)^2} \qquad \lambda_3 = \frac{-1}{(a_3)^2}$$

$$I = \frac{1}{(a_1)^2} + \frac{1}{(a_2)^2} + \frac{1}{(a_3)^2}$$



5. Paraboloid eliptic

ec. canonica: $$\frac{x^2}{(a_1)^2} + \frac{y^2}{(a_2)^2} = \frac{z}{a_3}$$

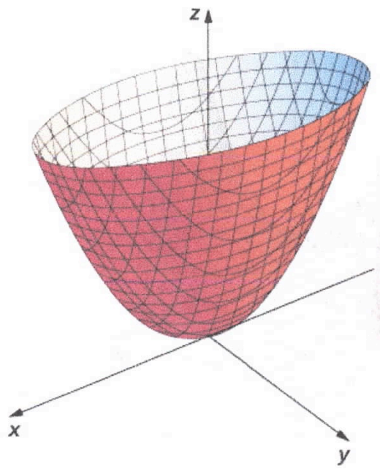
$$q(x) = \frac{x^2}{(a_1)^2} + \frac{y^2}{(a_2)^2} - \frac{z}{a_3} = 0$$

$$a = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \delta = 0$$

$$A = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 & 0 \\ 0 & \frac{1}{(a_2)^2} & 0 & 0 \\ 0 & 0 & 0 & \frac{-1}{2a_3} \\ 0 & 0 & \frac{-1}{2a_3} & 0 \end{pmatrix} \qquad \Delta = \frac{-1}{4(a_1)^2(a_2)^2(a_3)^2}$$

$$\lambda_1 = \frac{1}{(a_1)^2} \qquad \lambda_2 = \frac{1}{(a_2)^2} \qquad \lambda_3 = 0$$

$$I = \frac{1}{(a_1)^2} + \frac{1}{(a_2)^2}$$



6. Paraboloid hiperbolic

ec. canonică: $\frac{z}{a_3} = \frac{x^2}{(a_1)^2} - \frac{y^2}{(a_2)^2}$

$$g(x) = \frac{x^2}{(a_1)^2} - \frac{y^2}{(a_2)^2} - \frac{z}{a_3} = 0$$

$$a = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 \\ 0 & \frac{-1}{(a_2)^2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad \delta = 0$$

$$A = \begin{pmatrix} \frac{1}{(a_1)^2} & 0 & 0 & 0 \\ 0 & \frac{-1}{(a_2)^2} & 0 & 0 \\ 0 & 0 & 0 & \frac{-1}{2a_3} \\ 0 & 0 & \frac{-1}{2a_3} & 0 \end{pmatrix} \qquad \Delta = \frac{1}{4(a_1)^2(a_2)^2(a_3)^2}$$

$$\lambda_1 = \frac{1}{(a_1)^2} \qquad \lambda_2 = \frac{-1}{(a_2)^2} \qquad \lambda_3 = 0$$

$$I = \frac{1}{(a_1)^2} - \frac{1}{(a_2)^2}$$